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Are there neurotoxic effects of low-level carbon monoxide exposure? A systematic review of animal model studies

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Abstract

Background: Carbon monoxide (CO) is ubiquitously present in the environment and in the body. At high concentrations, it is known to cause severe neurotoxic effects both behaviourally and histologically, however the impact of low-level CO exposure, especially during prenatal development, remains unclear. This thesis aims to systematically review and synthesize available research in mammalian models to assess the behavioural and neurological effects of low-level CO exposure.

Methods: A systematic review was conducted to evaluate the literature examining the neurotoxic effects of CO exposure in mammalian models. A meta-analysis was subsequently performed on data extracted from seven studies that focused on behavioural outcomes of low-level CO exposure during pregnancy. The primary outcomes assessed were behavioural deficits associated with varying concentrations and durations of CO exposure.

Results: After full text screening, 207 papers were eligible for inclusion in the systematic review. Of these 207 papers, rats were the most commonly studied species with 148 papers studying this animal model in particular. Most research focused on adult exposure to CO (n=164). Within the subset of 28 papers which studied exposure to CO during pregnancy, the mean duration of exposure to CO was 400.7 hours, and the mean concentration of exposure to CO was 139.7 ppm.

A meta-analysis was conducted on the subset of papers with exposure to CO during pregnancy. The meta-analysis revealed a significant negative effect of low-level prenatal CO exposure on behavioural outcomes, with an overall effect size of -36.15% impairment in behavioural measures compared to controls (95% CI: -53.59 to -18.70, $p < 0.0001$). Stratified analyses indicated that certain CO concentrations - particularly 112.5 ppm and 225 ppm-, and durations - particularly 1.2 hours and 27 hours of exposure - were associated with significant behavioural deficits. However, no consistent dose-response relationship was observed across all exposure levels and durations. Meta-regression showed no significant relationship between CO concentration and effect size, suggesting that other factors may contribute to the observed variability.

Conclusions: The findings suggest that even low-level prenatal CO exposure can result in significant behavioural impairments in mammalian models. While specific concentrations of CO were found to have more pronounced effects, the lack of a clear dose-response relationship indicates

that other factors may also play a role in determining the severity of neurotoxic outcomes. These results underscore the need for further research, particularly studies that explore the molecular mechanisms underlying these effects and the potential influence of sex differences.

Acknowledgments

I would like to express my deepest gratitude to my supervisor, Dr. Robert Dickinson, for his invaluable guidance, insightful feedback, and unwavering support throughout the course of this project. His expertise and encouragement have been instrumental in shaping this research, and I am deeply appreciative of his mentorship.

I would also like to extend my heartfelt thanks to my friend and fellow master's student, Sara Arybou, whose constant support and friendship have kept me motivated during the more challenging moments of this journey. Her positivity and understanding have been a source of strength, and I am grateful to have shared this experience with her.

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List of Acronyms

SD Standard Deviation

CO Carbon monoxide

SEM Standard Error of the Mean

COHb carboxyhaemoglobin

DNS Delayed Neurologic Sequelae

H2S hydrogen sulfide

NO nitric oxide

REML restricted maximum likelihood

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1 Introduction

Carbon monoxide (CO) is part of a trinity of endogenous gasotransmitters alongside hydrogen sulfide (H2S) and nitric oxide (NO)[1]. At high exogenous concentrations, CO can be poisonous, leading to a range of symptoms and eventually leading to hypoxia and death if exposure remains high for an extended period of time [2]. CO poisoning leads to approximately 246 non-fire related hospital admissions and 4000 accident and emergency consultations in the UK alone every year [3]. It is mainly due to the fact that CO has a 200-fold affinity for hemoglobin, outcompeting oxygen and resulting in hypoxic stress [2], [4] . It has also been found to activate NMDA neurons - leading to a molecular cascade eventually resulting in lipid peroxidation [5], [6]. Given that the brain is especially vulnerable to ischaemic injury, it is in theory especially prone to damage during hypoxic stress [7].

Brain damage and behavioural deficits have indeed been observed in case studies of patients with carbon monoxide poisoning [8], who often present with Delayed Neurologic Sequelae (DNS) including personality changes, cognitive impairments and psychosis days to months after CO exposure [9], [10]. These neurological deficits of acute, high-level CO exposure are well documented [11].

At lower levels of CO, these neurocognitive effects are less well-known. CO is produced by the body to maintain normal physiology including cardiovascular and nervous system function - in which it is especially important for enabling and restoring synaptic plasticity in the brain [12] and vasodilation in the heart [13]. For this reason, it has been investigated for its pharmacologically therapeutic benefits in organ transplantation, pulmonary hypertension and malaria, for instance [14].

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One of the first to investigate the neurotoxic effects of lower levels of CO were John Haldane and his son, who both did arithmetic problems at increasingly higher ambient concentrations of CO [9]. Though their sample size was just themselves and the deficits in arithmetic could also be linked to fatigue or other factors, their work laid the foundations for more rigorous studies such as Schulte [15] in which volunteers were given psychological tests before and after exposure to CO, revealing that cognitive impairments occurred at carboxyhaemoglobin (COHb) levels (percentage of haemoglobin bound to CO) as low as 5%. This was confirmed by further research which demonstrated significant cognitive impairments were shown to exist at COHb concentrations of 1%-10% [16]. However these results were followed by others which showed no impairment in behavior in participants with very similar COHb levels [17], [18]. Overall studies in humans on the neurotoxic effects of low-level CO exposure are inconclusive.

With an increased awareness of the effects of smoking and pollution on health, epidemiological studies began to be conducted on lower levels of CO, such as those by Morris et al. [13], [19], highlighting associations between ambient CO levels and health outcomes, suggesting broader implications for public health. In the UK, these levels of ambient CO are quite common and can easily be found in non-ventilated rooms with radiators, for instance [9]. Despite being frequently overlooked, these low-level exposures are potentially insidious, suggesting a much larger burden of disease than previously thought. It is important to note that public health research looks at general trends in populations but can't control for other factors that might influence neurobehavior including other pollutants that may co-occur with CO in the air.

Today, it is widely accepted that CO at high levels can cause acute and chronic neurological, neurodevelopmental and behavioural damage [20]. Key studies over the decades have suggested that even low-level exposures can lead to significant neurological effects, however we currently lack a synthesis of the neurotoxic effects of low-level CO exposure in animal models and preclinical models.

The objectives of this systematic review are to:

- Evaluate the existing literature on the neurotoxic effects of low-level CO exposure in mammalian models.
- Conduct a meta-analysis on data regarding the neurological outcomes associated with such exposures, with an emphasis on behavioural outcomes

2 Methodology

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2.1 Protocol Development

A protocol outlining the methodology for this systematic review was developed and is available in Appendix C. The protocol details the research questions, inclusion and exclusion criteria, data extraction process, and analysis plan.

The search terms inputted into PubMed and EMBASE are as follows:

(”carbon monoxide”) AND (”low level*” OR “low dose*” OR “low concentration*” OR “chronic” OR “long term” OR “small dose*” OR “subacute” OR “toxic*” OR “poisoning” OR “pathophysiology” OR “pathological” OR “longer exposure*” OR “long exposure*” OR “lower level*” OR “injury” OR “deficits” OR “prolonged exposure” OR “sub-clinical” OR “acute” OR “brain”) AND (”preclinical” OR ”animal*” OR ”mammal*” OR ”rodent*” OR ”mice” OR ”mouse” OR ”rats” OR ”rat” OR ”rabbit*” OR ”monkey*” OR ”primate” OR ”primates” OR “swine” OR “porcine” OR “pig”)

2.2 Search Strategy

Literature published in English was considered, with searches conducted on Ovid Medline (PubMed) and EMBASE. The searches were performed independently on each database to ensure broad coverage of relevant literature. The search results from each database were downloaded and exported as a .csv file for further processing. Both of these files were imported into R and stitched together into one database using the code in Appendix A.

Deduplication was conducted by removing duplicate DOIs, abstracts, and titles using R code detailed in Appendix A.

2.3 Screening Process

The deduplicated file was imported into Excel for manual screening of titles and abstracts. Studies eligible for the literature review included peer-reviewed research with information on the neurotoxic effects of low-level carbon monoxide exposure which occurred during adult, juvenile and pregnant life stages. Eligible studies were published in English and were included regardless of the year of publication, initiation and duration of CO exposure or severity of the deficit.

Predefined criteria for exclusion include if the study only investigated human or in vitro models, no investigation of pure CO exposure, no outcome measures related to the neurotoxic effects of CO included or if the study investigated is a review article or a conference abstract

Two columns were used for exclusion criteria: one indicating whether a paper is to be excluded and another specifying the reason for exclusion.

Additionally, the animals' developmental stage and species in the studies were recorded in a separate column. Developmental stages were defined as follows:

- Pregnant: Exposed to carbon monoxide during pregnancy
- Neonate: Exposed to carbon monoxide from birth until 2 weeks old
- Juvenile: Exposed to carbon monoxide from 2 weeks old to 6 weeks old
- Adult: exposed to carbon monoxide after 6 weeks of age

The Excel sheet was then exported back to R for further filtering based on inclusion criteria. Reasons for exclusion after full-text screening were documented and the final database of 207 papers was saved using the code in Appendix A.

At this stage, additional information including the concentration of CO in ppm, species, sex, life stage, and general conclusions of the article were recorded.

Due to time constraints, the dataset was further subsetted to include only papers with exposure to CO in pregnant animals using code detailed in Appendix A.

2.4 Data extraction for meta-analysis

For each paper in the subset focusing on pregnant animals, additional columns were created to log whether histological and behavioural outcomes were measured. Of the 28 papers, 22 used histological measures and 10 used behavioral measures. Data extraction was then conducted on the 10 papers with behavioural measures.

Data was extracted by myself. Numerical or text data were extracted manually. Where this information was not directly available, graphical data were extracted using a digitizer tool (https://automeris.io/WebPlotDigitizer/). Data extraction focused on behavioural assessments. For each behavioural outcome, mean (X), standard error of the mean (Standard Error of the Mean (SEM)), standard deviation (Standard Deviation (SD)), and total number of animals per group (n) were extracted. Where SEM or SD were not available, their value was calculated as below:

$$SEM = \frac{SD}{\sqrt{n}}$$

Where the mean and SD were not provided, their values were approximated using the median, the sample size and the interquartile ranges based on the formulas from Luo, Wan, Liu, *et al.* [21] and Wan, Wang, Liu, *et al.* [22] respectively.

Additional metrics extracted for each experiment included the experiment type, the duration of exposure to CO in hours, the concentration of CO in ppm, and the sex of the animals studied.

Risk of bias was assessed using a modified version of the CAMARADES checklist, scoring each criterion from 0 (high risk of bias) to 9 (low risk of bias). (See Liang, Ahmad, and Dickinson [23] for more details on the modification of the checklist) [24], [25]

2.5 Meta-analysis

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A meta-analysis was conducted on the subset of pregnant animal data using normalized mean difference (NMD) as a measure of effect size because NMD is especially useful in animal studies where the out. All raw data were transformed into a format compatible with the CAMARADES meta-analysis web-based application (see user guide, https://bit.ly/3EB4mFX).

NMD was calculated according to the formula below: $NMD = \frac{\mu_c - \mu_{treatment}}{s_p}$

For this, the mean of the group treated with CO as μ_t, the mean of the control group as μ_c and the pooled standard deviation as

$$sp = \sqrt{\frac{(n_c-1)*SD_c^2+(n_t-1)*SD_t^2}{n_c+n_t-2}}$$

with SD_t corresponding to the standard deviation of the group treated with CO and SD_c corresponding to the standard deviation of the control group.

Variance for the NMD was calculated as:

$$\frac{n_c+n_t}{n_c*n_t} + \frac{NMD^2}{2(n_c+n_t)}$$

Control groups often served multiple treatment groups; thus, the 'true number of control animals (nc)' was calculated as below:

$$nc = \frac{number of controls}{number of experimental groups}$$

Pairwise and stratified meta-analyses were conducted through the CAMARADES meta-analysis web application (https://camarades.shinyapps.io/meta-analysis-app/). Individual effect sizes were weighted using the inverse variance method, which involves weighting each study by the inverse of its squared standard error ($1/se^2$)

For the meta-analysis, a random effects model was applied to the normalized mean differences (NMDs). The weighting was performed using the inverse variance method. The restricted maximum-likelihood estimator was used to calculate τ^2. The confidence intervals (CI) for τ^2 and τ were calculated with the Q-profile method.

Potential sources of heterogeneity were investigated through stratified meta-analyses. These predefined sources included the concentration of CO in parts per million (ppm), the duration of exposure to CO in utero in hours, the sex of the animals studied and the type of behavioural test used. The animal model used was considered a potential source of heterogeneity, however, all

studies included in this analysis used the same animal model. Differences between subgroups in the stratified meta-analyses were tested using a χ^2 test.

Egger's regression and funnel plots were constructed to assess between-study heterogeneity. The influence of funnel plot asymmetry on the overall effects was evaluated using the trim-and-fill method.

A meta-regression was conducted to explore the relationship between carbon monoxide concentration (measured in ppm) and the effect sizes reported across the included studies. This analysis was implemented to determine whether variations in CO concentration could explain the heterogeneity in effect sizes observed in the systematic review. The CO concentration was treated as a discrete moderator variable, categorized into specific levels (112.5 ppm, 150 ppm, 225 ppm, and 1000 ppm). The meta-regression analysis was performed using a mixed-effects model, which accounts for both within-study and between-study variability. The restricted maximum likelihood (REML) estimator was employed to estimate the model parameters, specifically the amount of residual heterogeneity (τ^2) and the overall effect of CO concentration on the effect sizes. The model included the intercept, which represents the baseline effect size when CO concentration is zero, and individual coefficients for each CO concentration level. The significance of the CO concentration levels was tested using the QM statistic, while the QE statistic was used to assess the presence of residual heterogeneity after accounting for the moderators. The τ^2 statistic provided an estimate of the residual heterogeneity after accounting for CO concentration, while I^2 and H^2 statistics were used to quantify the proportion of variability due to between-study differences versus sampling error. The overall R^2 value indicated the proportion of variability in effect sizes explained by the CO concentration levels.

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3 Results

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3.1 Literature review

Our search strategy specified in Figure 3.1, identified a total of 207 studies for systematic review. Languages of the papers are summarised in the table below. 68 papers were written in non-English languages without a translation and were excluded at the title and abstract screening stage (Table 3.1)

Data extraction was initially conducted on the 207 papers included to collect information about the species and the developmental stage (pregnancy, neonatal, juvenile or adult - see Methodology 2 for more information).

Most papers identified focused on adult animals. Of these 207, 164 papers studied adult animals, 28 studied pregnant animals, 2 studied neonatal animals and 13 studied juvenile animals. (Figure 3.1)

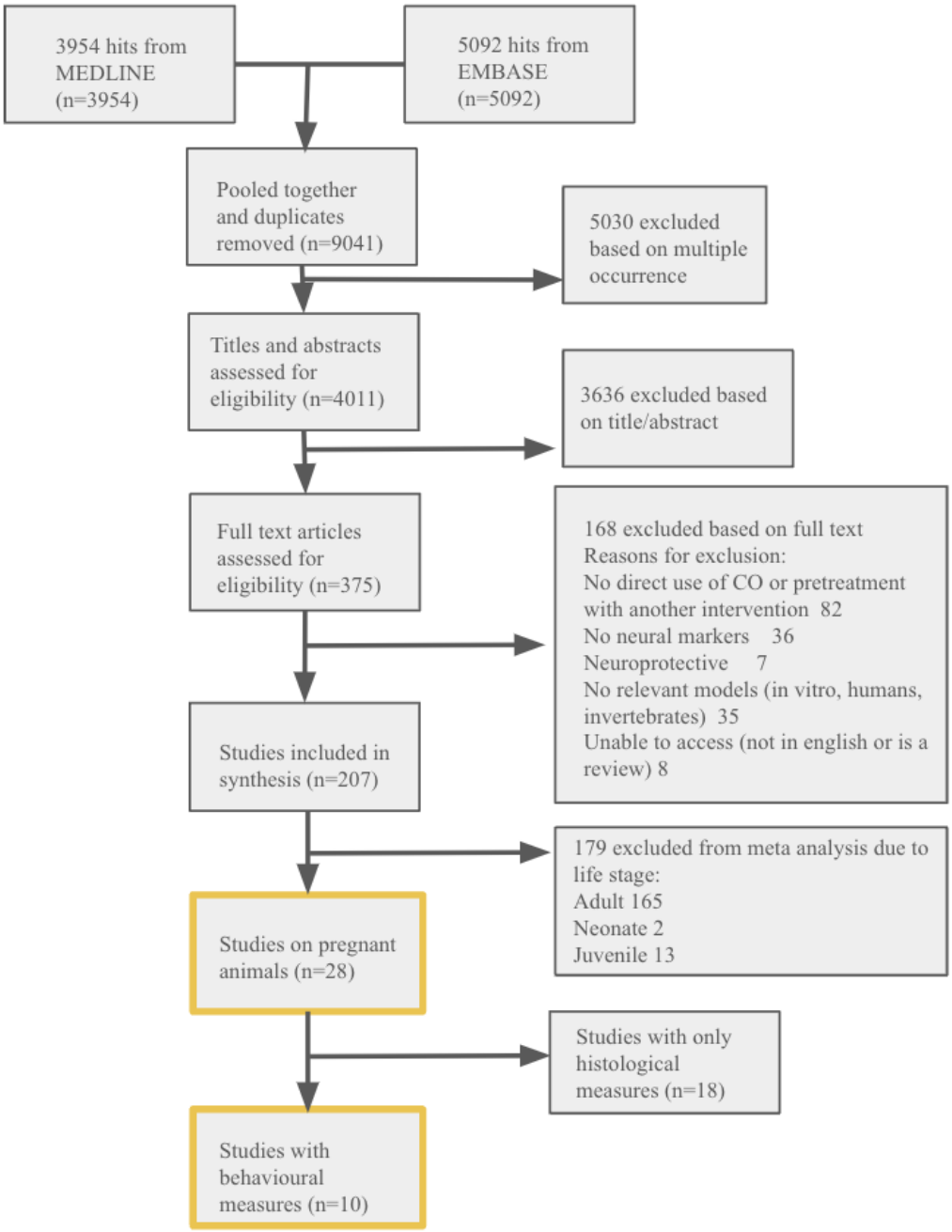


Figure 3.1: **PRISMA Flow Diagram for Study Selection Process.** PRISMA diagram detailing the systematic review process, including study identification, screening, and inclusion criteria. The initial database search yielded 9,046 hits, which were subsequently pooled and deduplicated to 4,011 unique studies. Titles and abstracts were assessed for eligibility, leading to the exclusion of 3,636 studies. Full-text articles were further reviewed, with 207 studies meeting the inclusion criteria. Of these, 28 studies focused on pregnant animals, and 10 of these examined behavioural outcomes, which were ultimately included in the meta-analysis due to time constraints. Exclusion reasons at each stage are detailed within the flowchart. Studies focusing on pregnant animals with behavioural measures, highlighted in yellow, were included in the final synthesis.

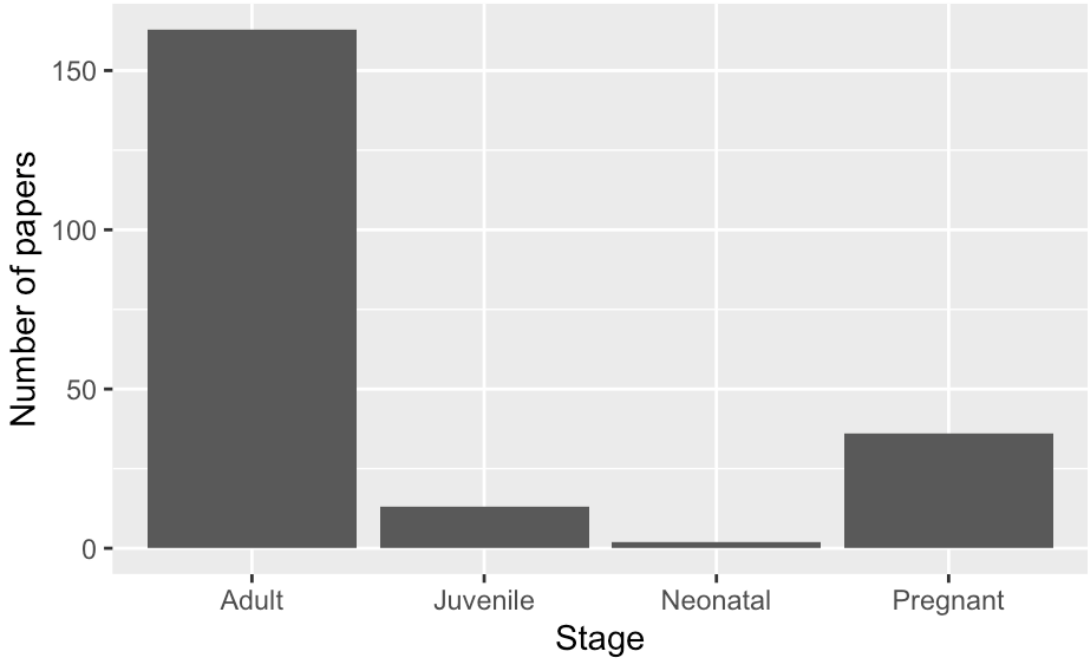


Figure 3.2: **Distribution of Included Studies by Developmental Stage** Bar chart illustrating the distribution of the 207 eligible studies included in the systematic review, categorized by the developmental stage of the animals studied: adult, juvenile, neonatal, and pregnant. The chart reveals a predominant focus on adult models (n = 165), with fewer studies investigating pregnant (n = 28), juvenile (n = 13), and neonatal (n = 2) animals. This distribution underscores the relative scarcity of research on CO exposure during early developmental stages, particularly in neonatal and juvenile animals.

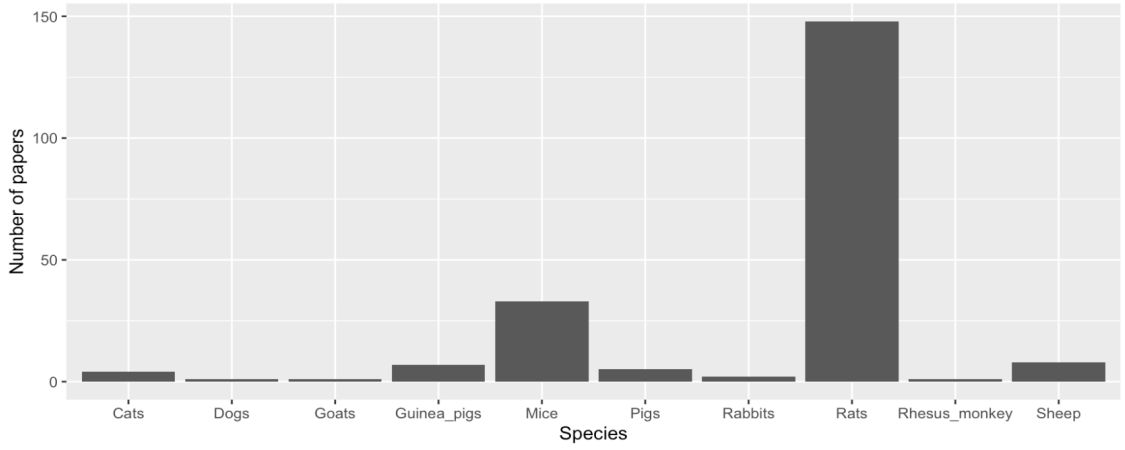


Figure 3.3: **Distribution of Included Studies by Animal Species** Bar chart showing the distribution of the 207 studies included in the systematic review based on the animal species used. The chart indicates that rats were the most commonly studied species, with 148 studies focused on this model. Mice were the next most studied species, followed by smaller numbers of studies on pigs, guinea pigs, and rhesus monkeys. Fewer studies involved other species, such as cats, dogs, goats, rabbits, and sheep. This distribution highlights the predominant reliance on rodent models, particularly rats, in the research on CO exposure, and suggests a potential gap in the exploration of its effects in non-rodent species.

Language	Counts
Chinese	37
English	4028
English, French	2
English, German	1
English, Spanish	1
English, Turkish	3
French	6
German	17
Italian	1
Japanese	7

Table 3.1: Language distribution of search results

Most papers focused on rats (n=148), with 33 papers investigating the neurotoxic effects of low-level CO in mice and the remaining papers studying cats, dogs, goats, guinea pigs, pigs, rabbits, rhesus monkeys and sheep.

3.1.1 Narrative review of the papers focused on pregnant animals

Within the subset of papers focused on pregnant animals, the sex of the surviving pups, duration of exposure to CO and concentration of CO in ppm were extracted. The mean CO concentration was 139.7ppm, with a median and IQR of 150ppm and 125ppm (Figure 3.4). The mean duration of exposure was 400.7 hours with a median and IQR of 424 and 72 (Figure 3.4). A qualitative summary of each paper in this subset was written (Appendix B).

After assessment with the modified CAMARADES risk-of-bias checklist, no high risk of bias, low quality (scores 1–3) studies were noted. 1 study (3%) was low risk of bias, high-quality (scores 7–9), and 27 (97%) of the studies assessed were identified as moderate quality, moderate risk of bias (scores 4–6). (Table 3.2)

Risk of bias	**Counts**
High (scores 1-3)	0
Moderate (scores 4-6)	27
Low (scores 7-9)	1

Table 3.2: Risk of bias Scores

Of these papers focused on pregnant animals, twenty-two papers were identified to include histological data, and ten papers included behavioural data (see Appendix B for a descriptive study of all the papers which had exposure to CO during pregnancy).

The ten papers which studied pregnant animals and focused on behavioural outcomes were selected for further analysis because of the time constraints of the project. Of these ten papers, seven papers had sufficient data for meta-analysis (see Figure 3.1 for a flowchart of the subset of data selected for quantitative meta-analysis).

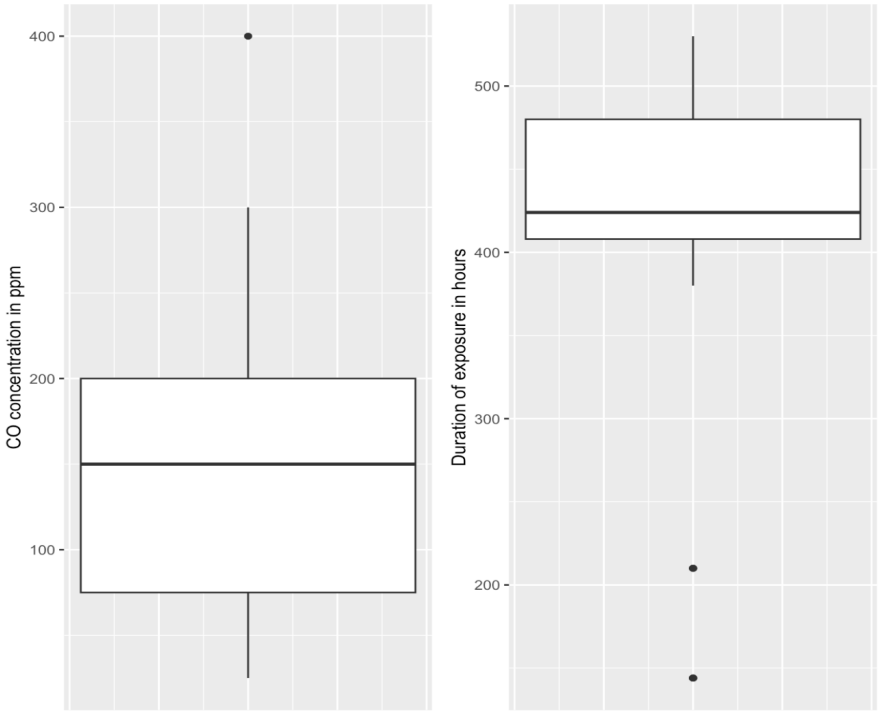


Figure 3.4: **Boxplots of CO Exposure Concentration and Duration in Utero** Boxplots illustrating the distribution of carbon monoxide (CO) exposure parameters among the studies included in the meta-analysis. The panel on the left shows the distribution of CO concentration in parts per million (ppm), with the median concentration around 150 ppm. The single study which investigated the effects of 1000ppm was omitted from the plot. The interquartile range (IQR) is represented by the top and bottom faces of the box, and the range is represented as the line emerging from the box. The panel on the right presents the distribution of the duration of CO exposure in utero, with a median exposure duration of about 424 hours and a broader range from 144 to 530 hours. These boxplots highlight the variability in both concentration and duration of CO exposure across the studies analyzed, which may contribute to differences in observed outcomes.

3.2 Meta-analysis

3.2.1 CO exposure significantly influences behavioural outcomes

A more in-depth phase of data extraction was conducted to include the mean and the standard error of the mean of the behavioural measures for each experimental group and each level of CO exposure for each of the studies. In total, 7 studies examined the behavioural effects of low-level carbon monoxide exposure. The meta-analysis combined results from k = 7 studies examining the impact of prenatal carbon monoxide (CO) exposure on behavioural responses. The overall effect size was estimated using a random effects model, which yielded a significant pooled effect size of -36.2 (95% CI [-53.6; -18.7], $p < 0.0001$), with no significant heterogeneity among the studies ($\tau^2 = 180.9$, $\tau = 13.5$, $I^2 = 32\%$, Q=8.8) (Figure 3.5).

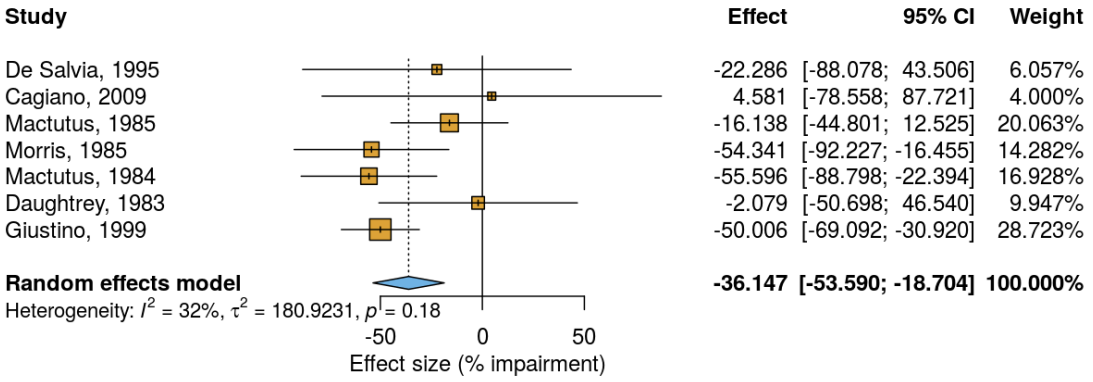


Figure 3.5: **Forest Plot of CO Exposure Impact on Behavioral Outcomes** Forest plot comparing effect sizes (with 95% confidence intervals [CIs]) for behavioural outcomes across seven studies investigating prenatal CO exposure. Each study is represented by a square, where the size of the square corresponds to the study's weight in the meta-analysis. Horizontal lines indicate the 95% CIs for each study's effect size. The overall effect size from the random effects model is shown as a blue diamond at the bottom, with the width representing the 95% CI. The overall pooled effect size is -36.2 (95% CI: -53.6 to -18.7, p $<$ 0.0001), with moderate heterogeneity (I^2 = 32%). This plot highlights both individual study contributions and the overall negative impact of CO exposure on behavioural outcomes in animal models.

3.2.2 Assessing Heterogeneity and Publication Bias

Sources of heterogeneity including the sex of the animals, the types of behavioural tests used the specific concentration of CO in ppm and the duration of exposure to CO were explored using stratified meta-analysis.

Sex

These findings highlight a significant negative effect of low-level CO exposure on behavioural outcomes in studies that uniquely focused on male offspring and in studies that included both male and female offspring. In the male subgroup, the effect size was -30.5 (SE = 13.1, 95% CI: -56.1 to -4.94, p = 0.0194), with moderate heterogeneity (I^2 = 39.7%, Q = 4.971, tau^2 = 281.4). In the mixed group (both males and females), the effect size was -41.5 (SE = 15.2, 95% CI: -71.4 to -11.7, p = 0.0064), also with moderate heterogeneity (I^2 = 44.6%, Q = 3.611, tau^2 = 295.1). A chi-square test for subgroup differences indicated no significant difference between the male alone and male and female combined groups (Q = 0.245, df = 1, p = 0.621) (Figure 3.6).

Group	ES	ci.low	ci.up	No.Animals	No.Studies
Male	-30.53	-56.11	-4.94	761	4
Both	-41.53	-71.38	-11.68	511	3
Global	-36.15	-53.59	-18.7	1272	7

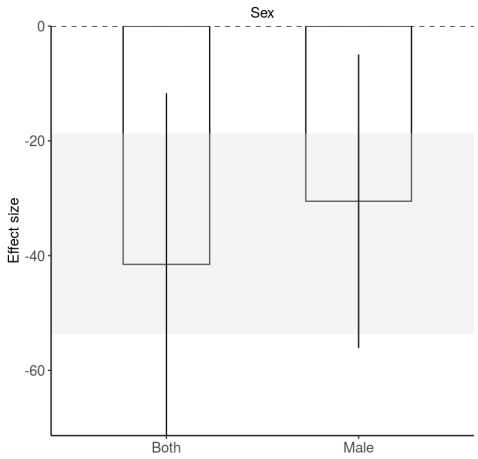


Figure 3.6: **Barchart of Effect Sizes by Sex Group** Barchart illustrating the effect sizes of CO exposure on behavioural outcomes, categorized by sex group: male-only studies, studies including both males and females and the overall global effect (shaded in grey). Lines through the bars symbolize upper and lower confidence intervals. The upper table summarizes the effect size (ES), 95% confidence intervals (CI), number of animals, and the number of studies for each group. The bar chart shows that studies including both sexes exhibited a more negative effect size (-41.5), with a wider CI compared to male-only studies (-30.5). The global effect size, derived from all included studies, was -36.1. This visualization indicates that sex may influence the behavioural impact of CO exposure, although the overlapping confidence intervals suggest that the difference may not be statistically significant.

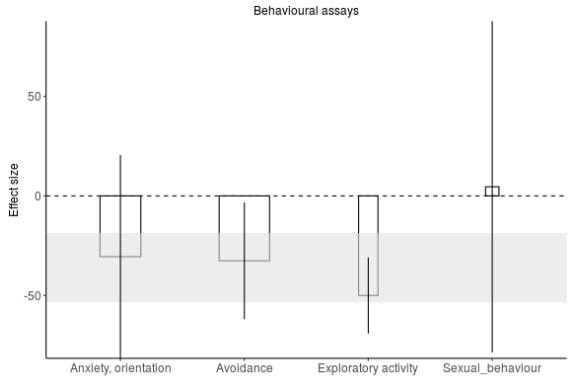


Figure 3.7: **Barchart of Effect Sizes Across Different Behavioral Assays** Barchart showing the effect sizes of CO exposure on various behavioural assays, including anxiety and orientation, avoidance behaviour, exploratory activity, and sexual behaviour. The effect sizes indicate varying impacts of CO exposure, with exploratory activity showing the most significant negative effect. The shaded area represents the range of effect sizes observed across all studies. The plot demonstrates that CO exposure has the potential to negatively affect a range of behaviours, with the most consistent impairments observed in exploratory activity and avoidance behaviour. The small sample size for sexual behaviour assays is reflected in the wide confidence intervals, indicating higher variability and less reliable estimates for this category.

Behavioural test used

A chi-square test for subgroup differences between the different types of behavioural tests used indicated no significant difference between the subgroups (Q = 2.86, df = 3, p = 0.413), suggesting that the behavioural impact of CO exposure is consistent across different types of behavioural outcomes. In the avoidance behaviour subgroup, the effect size was -32.6 (SE = 15.0, 95% CI: -61.9 to -3.25, p = 0.0295), with moderate heterogeneity (I^2 = 37.5%, Q = 3.20, τ^2 = 284). In the exploratory activity subgroup, the effect size was -50.0 (SE = 9.74, 95% CI: -69.1 to -30.9, p < 0.0001), with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA). Similarly, for the anxiety and orientation subgroup, the effect size was -30.5 (SE = 26.0, 95% CI: -81.5 to 20.5, p = 0.2409), showing high heterogeneity (I^2 = 63.8%, Q = 2.76, τ^2 = 871.2). However, the sexual behaviour subgroup showed no significant effect (effect size = 4.581, SE = 42.4, 95% CI: -78.6 to 87.7, p = 0.914), with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA) (Figure 3.7)

Impact of different concentrations of CO

The third subgroup analysis evaluated the impact of CO exposure across different dosages, revealing significant negative effects for several dosage groups (Table 3.3). However, only two CO concentration groups reached significance. In the 112.5ppm dosage group, the effect size was -47.9 (SE = 9.35, 95% CI: -66.2 to -29.524, p < 0.0001), with no heterogeneity reported (I^2 = 0%, Q = 0.629, tau^2 = 0.000). In the 225ppm dosage group, the effect size was -54.3 (SE = 19.330, 95% CI: -92.2 to -16.5, p = 0.0049), also with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA).

For the other concentrations, the effect size was not found to be significant. In the 150ppm group, the effect size was -34.9 (SE = 19.7, 95% CI: -73.6 to 3.685, p = 0.0762), showing high heterogeneity (I^2 = 67.8%, Q = 3.12, τ^2 = 528.1). The 100ppm dosage group showed no significant effect (effect size = 4.58, SE = 42.4, 95% CI: -78.6 to 87.7, p = 0.914), with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA). Similarly, the 1000 ppm dosage group did not show a significant effect (effect size = -2.079, SE = 24.8, 95% CI: -50.7 to 46.5, p = 0.933), with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA).

A chi-square test for subgroup differences indicated no significant difference between the CO groups (Q = 5.089, df = 4, p = 0.278), suggesting that there is a consistent behavioural deficit across different concentrations of CO.

CO concentration	ES	ci.low	ci.up	No. Animals	No. Studies
112.5	-47.85	-66.18	-29.52	145	2
100	4.58	-78.56	87.72	48	1
150	-34.94	-73.57	3.68	634	2
225	-54.34	-92.23	-16.45	208	1
1000	-2.08	-50.7	46.54	237	1
Global	-36.15	-53.59	-18.7	1272	7

Table 3.3: Summary of effect sizes (ES) with confidence intervals (ci.low, ci.up), number of animals, and number of studies for different CO concentrations.

Impact of different durations of exposure to CO

Different durations of exposure to CO did not show significant differences in effect size either (Q=7.25, df=5, χ^2=11.07, p=0.202). For the 1.2-hour exposure group, the effect size was -54.3 (SE = 19.3, 95% CI: -92.2 to -16.5, p = 0.0049), with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA). Significant effect sizes were identified for the studies with exposure duration averaging 26.72 hours, with an effect size of -55.6 (SE = 16.9, 95% CI: -88.8 to -22.4, p = 0.0010), also with no heterogeneity reported (I^2 = NA, Q = 0.000, τ^2 = NA). At 79.69 hours, 138.3 hours,

364 hours and 480 hours no significant effect was identified (effect sizes of -2.079±24.8, -16.1±14.6, -22.3±33.6 and -38.3±22.4 respectively) (Table 3.4)

Duration in hours	ES	ci.low	ci.up	No. Animals	No. Studies
1.2	-54.34	-92.23	-16.45	208	1
26.73	-55.6	-88.8	-22.39	66	1
79.69	-2.08	-50.7	46.54	237	1
138.3	-16.14	-44.8	12.52	568	1
364	-22.29	-88.08	43.51	45	1
480	-38.33	-82.2	5.55	148	2
Global	-36.15	-53.59	-18.7	1272	7

Table 3.4: Summary of effect sizes (ES) with confidence intervals (ci.low, ci.up), number of animals, and number of studies for different durations of exposure to CO.

Trim and fill analysis

Trim-and-fill analysis was used to identify funnel plot asymmetry. 2 imputed studies were suggested on the left side, as shown in the figure (Figure 3.8) (SE = 1.82). The estimated effect size when using the REML method and accounting for these two imputed studies remained significantly negative (-42.0%, SE=8.66, p<0.0001) with moderate heterogeneity (I^2=35.99%). The confidence interval does not cross zero (CI=[-58.95:-24.99]), suggesting a robust negative effect.

An Egger's regression test was conducted to further assess potential publication bias in the meta-analysis. The analysis revealed a significant association between precision and standardized effect size, with a coefficient of -61.94 (SE = 16.92, t = -3.661, p = 0.0146). This indicates that studies with higher precision (larger sample sizes) tend to report smaller effect sizes. The intercept was 1.47 (SE = 1.00, t = 1.469, p = 0.202), which was not statistically significant. The model explained 72.8% of the variance in effect sizes (R^2 = 0.728), with an adjusted R^2 of 0.674. These results suggest that there may be a presence of publication bias but that this does not significantly influence the overall outcome of this meta-analysis (Figure 3.9)

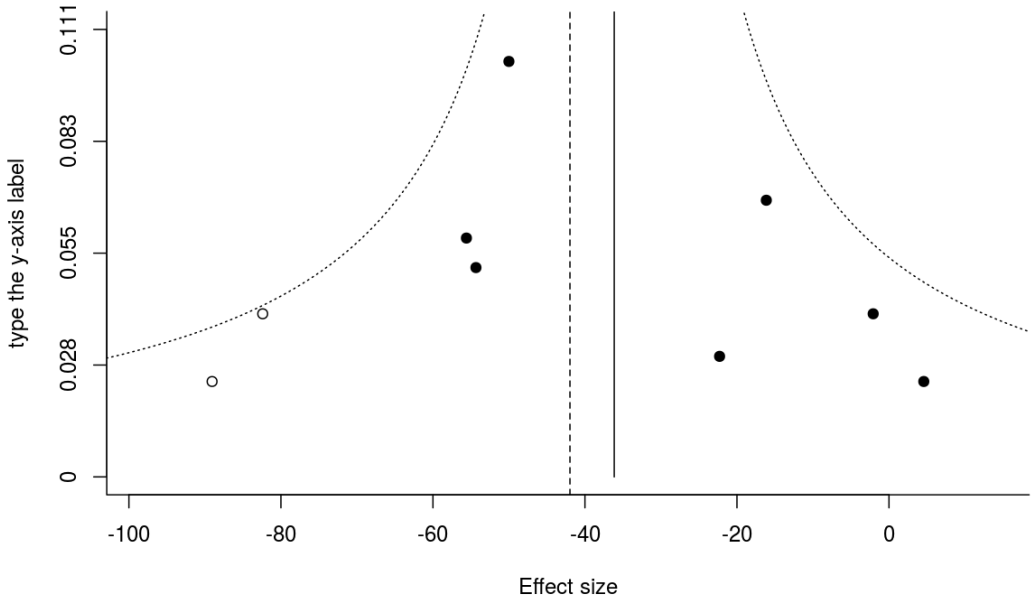


Figure 3.8: **Funnel Plot for Assessing Publication Bias.** Funnel plot evaluating the potential for publication bias in the meta-analysis of CO exposure studies. Each point represents a study, with effect sizes plotted against a measure of study precision (standard error). Empty points represent missing studies. The plot includes a solid vertical line representing the overall effect size, a dashed vertical line representing the potential effect size when the missing studies are accounted for and two curved dashed lines indicating the expected distribution of studies in the absence of bias. The asymmetry observed, particularly the clustering of studies on the left side of the plot, suggests the potential presence of publication bias, with smaller studies (those with higher standard errors) showing larger and more variable effect sizes. This pattern indicates that negative or non-significant results may be underreported or unpublished.

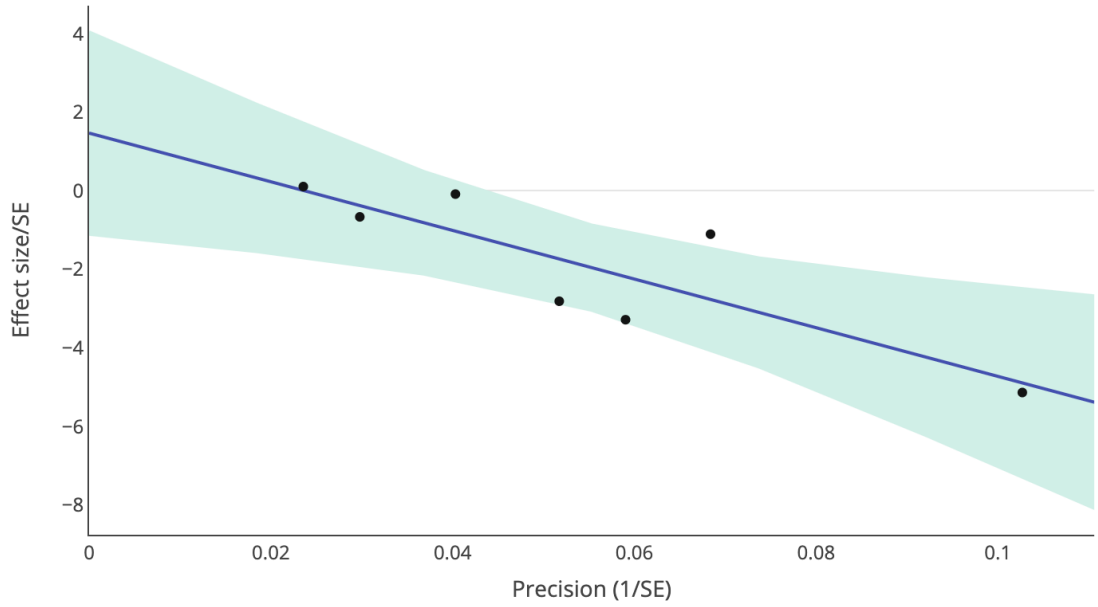


Figure 3.9: **Egger's Regression Plot for Assessing Publication Bias.** Egger's regression plot depicting the relationship between effect size (scaled by standard error (SE)) and study precision (1/SE) to assess potential publication bias in the meta-analysis. The plot shows a negative slope, indicating a possible small-study effect where smaller studies (lower precision) report larger effect sizes. The shaded area represents the 95% confidence interval around the regression line. The trend observed in this plot suggests that non-significant publication bias may be present, with more precise studies (larger sample sizes) tending to show smaller effect sizes, while less precise studies report more extreme effects.

3.3 Meta-regression

A meta-regression was conducted to assess the relationship between carbon monoxide (CO) concentration (ppm) and effect size across seven studies. The mixed-effects model, using the REML estimator, showed no residual heterogeneity (² = 0, SE = 695.403), with $I^2 = 0.00\%$ and $H^2 = 1.00$, indicating that all variability was due to sampling error. The test for residual heterogeneity was non-significant (QE(df = 1) = 0.0000, p = 1.0000), confirming that the model fully accounted for the observed heterogeneity ($R^2 = 100.00\%$).

However, the test of moderators (CO concentration levels at 112.5 ppm, 150 ppm, 225 ppm, and 1000 ppm) was not statistically significant (QM(df = 5) = 8.827, p = 0.116), indicating that CO concentration did not significantly influence effect size. The overall effect size was marginally significant (estimate = 1.0000, SE = 0.517, p = 0.0532), but individual CO levels did not produce significant effects (Figure 3.10).

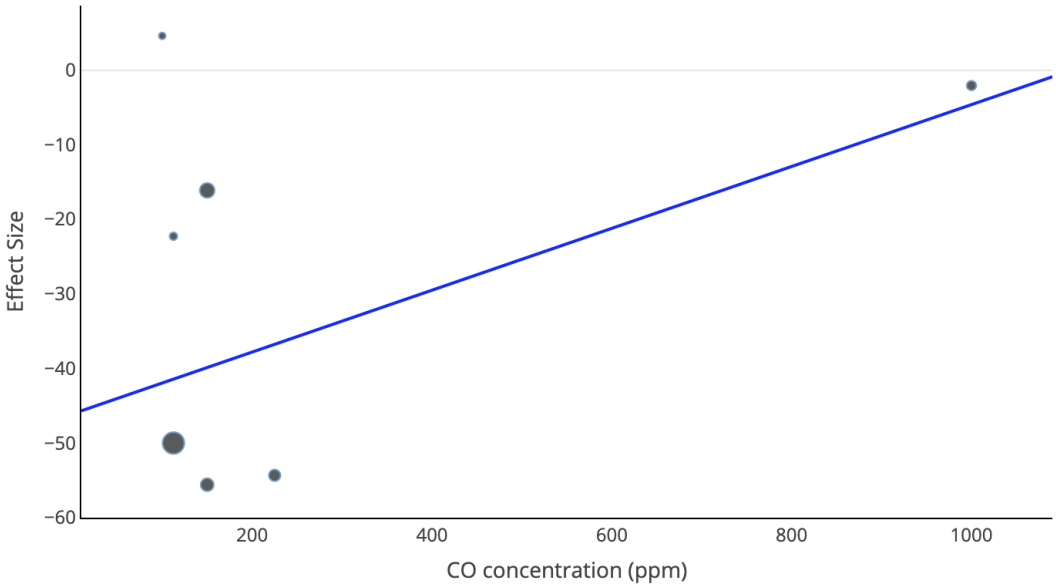


Figure 3.10: **Meta-regression plot of CO concentration (ppm) versus effect size** The plot illustrates the relationship between carbon monoxide concentration (ppm) and effect size across seven studies. The size of the points reflects the weight of each study in the meta-regression. The blue regression line indicates a positive trend, suggesting that higher CO concentrations are associated with larger (less negative) effect sizes, although the individual concentration levels did not significantly influence the effect sizes.

4 Discussion

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This systematic review and meta-analysis provide evidence regarding the neurotoxic impact of low-level CO exposure in animal models. Through an initial detailed examination of 207 studies, I then focused on a subset that investigated CO exposure during pregnancy, ultimately performing a meta-analysis on seven studies with sufficient data on behavioural outcomes (Figure 3.1).

4.1 Systematic review

Risk of bias The quality assessment of the 28 papers that investigated exposure to CO during pregnancy revealed that 97% of the studies were of moderate quality with moderate risk of bias (scores 4–6), while only one study was of high quality with low risk of bias (score 7–9) (Table 3.2). This distribution suggests a need for more rigorous research designs to improve the reliability of findings in this field.

Focus on animal models and type of animal model used While this systematic review focused on animal models to assess the effects of low-level CO exposure, it is important to acknowledge the limitations of this approach. The controlled settings of animal studies often do not account for the complexity of human environmental exposures, such as variations in lifestyle, and simultaneous exposure to other toxins. Given that most studies eligible for inclusion in the

systematic review focus on rats (n=148) and mice (n=33), the results of this review are further constrained by the limited amount of information available from non-rodent models. Animal models are valuable tools for understanding biological mechanisms and predicting potential effects in humans, but they are not necessarily accurate measures of human physiology and behaviour [26]. Given the difference between rodent brains and human brains in particular [27], this can lead to variations in how CO exposure impacts behavioural and neurological outcomes. Future reviews should aim to incorporate information from complementary approaches, including epidemiological research and more complex models, to provide a more comprehensive understanding of CO's impact on human health.

Exposure type Of the 207 studies, 164 focused on exposure to CO during adulthood. Only a minority of studies investigated the neurotoxic effects of CO exposure in pregnant, neonatal and juvenile animals. These results highlight disparities in research focus in the field and the need for more research, especially in non-adult models.

Sex bias in the literature It is important to note that of the 28 pregnancy papers, many of the studies focused primarily on male offspring. Though subgroup analysis of the effect of sex on the effect size was non-significant when comparing studies with only male animals and studies with both male and female animals, this gender bias in preclinical studies could limit the generalizability of the findings. In fact, there are known sex differences in physiological and behavioural responses to environmental toxins and gasotransmitters [28]. One experimental condition in one of the studies [29] compared female and male animals' responses to an avoidance task, and found that female offsprings' behavioural performance was significantly less affected by low-level CO exposure relative to their male counterparts'. In fish, there are also sex-specific effects of CO exposure [30]. Future research should aim to include both male and female subjects to provide a more comprehensive understanding of the effects of CO exposure during prenatal development in mammals.

4.2 Meta-Analysis

The meta-analysis demonstrated a significant negative impact of prenatal CO exposure on behavioural outcomes. The overall effect size was -36.147 (95% CI: -53.5901 to -18.7038, $p < 0.0001$), with moderate heterogeneity ($I^2 = 32\%$), suggesting that low-level CO exposure during pregnancy can significantly impair behavioural development (Figure 3.5). These results align with our fun-

damental understanding of the role of CO in pregnancy, where, along with NO and H_2S, it plays important roles in pregnancy adaptation and signalling for normal labour [31]. These findings also align with government recommendations that emphasize the increased vulnerability of pregnant individuals to CO exposure. The Centers for Disease Control and Prevention (CDC) and other health agencies have highlighted that pregnant individuals are particularly susceptible to the harmful effects of CO due to physiological changes during pregnancy, which can lead to greater absorption and slower elimination of CO, thereby posing significant risks to both the mother and the developing fetus [32]. Human studies have further supported these concerns, demonstrating that even low levels of CO exposure can lead to adverse outcomes, such as reduced birth weight, preterm birth, and developmental issues in children exposed in utero [33]–[35]. Overall these studies underscore the importance of stringent public health guidelines and targeted interventions to protect vulnerable populations from CO exposure, especially in environments where CO levels might be elevated due to poor ventilation, malfunctioning appliances, or environmental pollution. My research corroborates these findings and reinforces the need for continued vigilance in monitoring and mitigating CO exposure in at-risk populations.

My original hypothesis was that there may be a dose-response relationship between levels of CO exposure and the severity of behavioural outcomes. This is based on previous research which observed a threshold of deficit of fetal growth at 400ppm, but not at lower levels of CO [36] when the duration of exposure was held constant. This hypothesis is all the more plausible given that low levels of endogenous and exogenous CO are ubiquitously present, and toxicity of CO should therefore be concentration-dependent, because if it weren't, then even very small amounts of environmental exposure would be lethal. If this hypothesis were true in this analysis. duration of exposure and concentration of CO should be significant sources of heterogeneity. To test this, subgroup analyses of duration and concentration were conducted. The subgroup analysis of CO concentration revealed that there was no overall relationship between CO exposure levels and severity of behavioural outcomes, however, there are significant negative effects of CO exposure at specific concentrations. Notably, the 112.5 ppm and 225 ppm dosage groups exhibited substantial negative impacts on behavioural outcomes, with effect sizes of -47.85 (95% CI: -66.18 to -29.52, $p < 0.0001$) and -54.34 (95% CI: -92.23 to -16.45, $p = 0.0049$), respectively. These findings were observed with no heterogeneity reported ($I2 = 0\%$ for 112.5 ppm). Other concentration groups, such as 150 ppm, 100 ppm, and 1000 ppm, did not show significant effects. These results were confirmed in a subsequent meta-regression, which also showed that the effects of CO concentration on effect size were non-significant (p=0.052).

Duration of exposure in utero also did not influence the severity of the effect that CO exposure had on behaviour, however significant effect sizes were observed for 1.2 hours and for 27 hours of exposure in particular. This suggests that prolonged exposure does not necessarily exacerbate the neurotoxic effects of CO, which raises questions about the underlying factors that might contribute to this phenomenon. It is possible that other variables, such as the timing of exposure during critical periods of development, genetic predispositions, or compensatory mechanisms within the developing brain, play a more significant role in determining the extent of behavioural impairment. Altogether, the results of the subgroup analyses were surprising and indicate that there may not necessarily be a simple dose-response relationship between CO exposure and behavioural deficits, or that this relationship may be more complex than a simple linear one.

It is possible that other variables, such as the timing of exposure during critical periods of development, genetic predispositions, or compensatory mechanisms within the developing brain, play a more significant role in determining the extent of behavioral impairment. Additionally, the complexity of CO's interaction with various physiological systems may mean that its effects plateau after a certain threshold, rather than increasing linearly with longer exposure durations. Further research is needed to understand these factors and to elucidate the relationship between CO exposure and neurobehavioral outcomes.

The observed effect may also be due to factors such as the use of standardized levels of CO in the field. Indeed, many of the studies included in the meta-analysis used 75ppm and 150ppm as standardized levels of exposure, and therefore significance may have been achieved through the law of large numbers, whereas smaller studies using different concentrations might have not controlled for other factors as rigorously. However, this postulation would go against the findings from Egger's regression where smaller studies tended to show larger effect sizes, whereas larger studies showed more moderate effect sizes. It is also possible that the effects on behaviour are volume-dependent rather than specifically concentration or duration-dependent. That is to say that exposure to a higher concentration for a shorter duration might show fewer behavioural deficits than exposure to a lower concentration of CO with a longer duration, for instance. Perhaps future analyses can also compute the effect size of the product of concentration and duration as opposed to their addition.

Finally, the precise timing of CO exposure during gestation was not taken into account in this systematic review, which could be an important factor influencing the observed outcomes. For instance, studies have shown that exposure to CO during critical periods of brain development, such as during the late gestation period, can lead to more pronounced neurodevelopmental deficits compared to exposure at other times [34], [37]. Conversely, exposure at other stages of development

may have minimal or no effects, indicating that the timing of CO exposure is crucial in determining its impact. By carefully controlling and documenting the timing of CO exposure, future studies could identify windows of vulnerability during development and better understand the mechanisms through which CO exerts its effects. This has already been done using brain organoids to understand the neurodevelopmental impacts of arsenic and lead [38], but it would be interesting to see an adaptation of this protocol with an emphasis on CO exposure.

This systematic review serves as an exploratory investigation into the effects of low-level CO exposure on behavioural outcomes, laying the groundwork for future research. While my analysis provides valuable insights into the potential impact of CO, it primarily focuses on the observable behavioural deficits rather than the underlying molecular mechanisms. To gain a deeper understanding of how CO exerts its effects, future research should concentrate on the molecular basis of these outcomes. This could involve examining the role of CO in cellular signalling pathways, its interaction with neurotransmitters, and its impact on gene expression related to neurodevelopment and behaviour. Investigating these molecular processes will help elucidate the specific pathways through which CO exposure leads to behavioural changes and may identify potential biomarkers for early detection and intervention. By shifting the focus to the molecular level, future studies can provide more targeted insights, ultimately contributing to the development of more effective prevention and treatment strategies.

4

Conclusions

This systematic review and meta-analysis provide evidence that prenatal exposure to low levels of carbon monoxide (CO) can significantly impair neurobehavioral outcomes in mammalian models. My analysis identified specific concentrations of CO, particularly 112.5 ppm and 225 ppm, as having the most pronounced negative effects on behavior, with observed effect sizes indicating substantial impairments. However, the lack of a clear dose-response relationship and the variability in results across studies suggest that factors other than concentration alone—such as the timing of exposure, duration, and the specific behavioral assays used— may play critical roles in determining the extent of CO's neurotoxic effects.

The findings of this research align with existing concerns regarding the vulnerability of certain populations, such as pregnant individuals, to the detrimental effects of CO exposure. Despite the significant insights provided by animal models, it is crucial to acknowledge their limitations. The differences in physiology, metabolism, and environmental interactions between animals and humans underscore the need for caution when extrapolating these results to human populations. Future studies should strive to include more diverse animal models and in vitro models, focus on both sexes, and consider the molecular mechanisms underlying CO-induced developmental neurotoxicity.

Moreover, this research highlights the importance of controlling for the timing of CO exposure during prenatal development, as certain critical periods may be more susceptible to its harmful effects. Addressing these gaps in future research could lead to a more comprehensive understanding of CO's impact and inform public health strategies aimed at mitigating risks associated with low-level CO exposure.

Overall, these results suggest significant neurotoxic risks associated with low-level prenatal CO exposure, emphasizing the need for further investigation into the underlying mechanisms and the development of more targeted interventions to protect against these effects.

A

Code used for deduplication of files

Reading in the files with the search results from EMBASE and Medline and combining it into one large dataframe

```
library(dplyr)
embase<-dplyr::bind_rows(embase_sub2000, embase_2000)
all_bind<-dplyr::bind_rows(embase,medline) # at this stage we have 9041 papers
```

Deduplication of the files by removing rows with duplicate titles, abstracts and DOIs.

```
  all_bind_dedup_ti<-as.data.frame(distinct(all_bind, TI, .keep_all = TRUE))
  #removing rows with duplicate titles
all_bind_dedup_ab<-distinct(all_bind_dedup_ti, AB, .keep_all = TRUE)
#removing rows with duplicate abstracts
all_bind_dedup_do<-distinct(all_bind_dedup_ab, DO, .keep_all = TRUE)
#removing rows with duplicate doi
dedup_done<-cbind(all_bind_dedup_do$TI, all_bind_dedup_do$AB,
all_bind_dedup_do$AU, all_bind_dedup_do$PT) #this is a file with just the titles
and abstracts after deduplication. I can screen these manually.
#At this point we have 4011 to manually screen.
write.csv(dedup_done,"/Users/Coding/Downloads/deduped_simplified_2.csv")
```

Filtering data and quantifying reasoning for inclusion and exclusion after full text screening

```
complete <- read.csv("/Users/Coding/Downloads/whole_text(in).csv")
complete$Reason.to.exclude <- as.factor(complete$Reason.to.exclude)
exclusion <- count(complete$Reason.to.exclude)  # This counts how many papers were ex

keep <- subset(complete, complete$Remove == "no")  # Saved a file on my computer of a
removed <- subset(complete, complete$Remove == "yes")
write.csv(keep, "/Users/Coding/Downloads/whole_text_final.csv")
```

A

Subsetting the data into only papers which focused on exposure during pregnancy

```
species <- read.csv("/Users/Coding/Downloads/FULL TEXT SCREENING DONE(in1).csv")
Pregnant <- subset(species, Stage == "Pregnant")  # 28 observations
```

B

Summary of papers with exposure to CO during pregnancy

B

Authors	Title	Histology	Behaviour	Summary
Tolcos M. Mallard C. McGregor H. D.	prenatal carbon monoxide and postnatal hyperthermia: Short and long-term effects on neurochemicals and neuroglia in the developing	Cerebral Hemisphere, Cerebral cortex, Subcortical white matter, Thalamus, Basal ganglia, Hippocampus, Cerebellum, Medulla; Immunoreactivity for GFAP, 5-HT, ME, SP, and TH		Significant increase in GFAP-IR in several regions of the medulla at 8 weeks for the hyperthermia condition but not the normothermia condition. Lesions and gliosis observed at 8 weeks in the cerebral cortex, thalamus, and medulla. The increase in GFAP-IR is associated with hyperthermia exposure postnatally.
Cagiano R. Desantis S. Labate M. Tattoli M. Trabace L.	Prenatal exposure to low levels of carbon monoxide alters sciatic nerve myelination in rat offspring.	Computer-assisted morphometric analysis of sciatic nerve myelination	Motor activity measured by infrared sensors	Significant reduction in myelin sheath thickness was noted in older offspring (40 and 90 days) for both CO exposure conditions. No discernible effects on body weight compared to controls. The severity of myelin deficit was consistent with the level of CO exposure, with more pronounced effects at 150 ppm than at 75 ppm. Motor activity not significantly affected by CO exposure
Tolcos M. Mcgregor H. Walker D. Rees S.	exposure to carbon monoxide results in a reduction in tyrosine hydroxylase-immunoreactivity and an increase in choline acetyltransferase-	(spino-medullary junction to just below the cerebellar peduncles) and cerebellar weights; Immunoreactivity and counts for TH, ChAT, m2, SP, ME, and GFAP in medulla sections; Western blotting of tyrosine hydroxylase (TH) protein		Analysis of the medulla for neurochemical changes; significant changes in tyrosine hydroxylase and choline acetyltransferase immunoreactivity

Carratu M.R. Cagiano R. Tattoli M. Trabace L. Borracci P. Cuomo V.	Prenatal exposure model simulating CO inhalation in human cigarette smokers: Sphingomyelin alterations in the rat sciatic nerve.	Brains and sciatic nerves processed for analysis of free sphinganine and sphingosine concentrations;	Motor activity measured by infrared sensors	Sphinganine levels in the sciatic nerve did not change significantly compared to control. Sphingosine levels in the sciatic nerve were increased by 2.35-fold compared to control (statistically significant). As a result, the SA/SO ratio was significantly reduced in the sciatic nerve of CO-exposed rats compared to controls, indicating an alteration in lipid metabolism possibly due to hypoxia induced by CO. No changes in motor activity or motor skills.
Leffler C.W. Nasjletti A. Yu C. Johnson R.A. Fedinec A.L. Walker N.	Carbon monoxide and cerebral microvascular tone in newborn pigs.	Western blot analysis of heme oxygen-ase-1 (HO-1, A) and heme oxygenase-2 (HO-2, B) in cerebral vessels		CO application led to dose-dependent dilation of pial arterioles, significant at 10^-11 M in smaller arterioles and at 10^-10 M in larger arterioles, maximal by 10^-9 M; , abolished by KCa channel inhibitors like tetraethylammonium chloride and iberiotoxin. Responses to heme-L-lysinate (an HO substrate) mimicked those of CO, suggesting CO production via HO activity contributes to vascular effects. CO and heme-L-lysinate slightly increased cAMP but did not significantly affect cGMP levels in the CSF. Overall results confirmed that endogenous CO production can modulate vascular tone by affecting calcium-activated potassium channels, contributing to cerebral microvascular dilation in newborn pigs.

B

Giustino A. Cagiano R. Carratu M.R. Cassano T. Tattoli M. Cuomo V.	Prenatal exposure to low concentrations of carbon monoxide alters habituation and non-spatial working memory in rat offspring.		Novel object recognition	Across all groups (0 ppm, 75 ppm, and 150 ppm), there were no significant differences in spontaneous motor activity in the open field test, suggesting that the observed differences in cognitive tests were not due to changes in general activity levels. Habituation: CO-exposed groups failed to show habituation; they did not decrease their exploration upon repeated presentation, indicating a failure to recognize the objects as familiar. Impaired ability to discriminate between the novel and familiar objects at both CO concentrations, indicating deficits in working memory. Both CO-exposed groups (75 ppm and 150 ppm) showed significantly less exploration time during the first trial compared to the control group, suggesting a diminished interest or cognitive engagement with the objects. Control rats and 75 ppm exposed rats showed a decrease in exploratory activity from Trial 1 to Trial 2, which is typical for habituation. In contrast, 150 ppm exposed rats did not show a significant decrease, indicating impaired habituation. Control rats spent significantly more time exploring the novel object than the familiar one, demonstrating proper memory function. However, both CO-exposed groups did not show a significant preference for the novel object, indicating a failure in memory
Ancona D. Cassano T. Tattoli M. Trabace L.	exposure to low concentrations of carbon monoxide on sexual behaviour and mesolimbic dopaminergic function in rat offspring.		behaviour by analysis of video recordings; extracellular concentrations of DA, DOPAC and HVA in the NAC by	No significant differences in locomotor activity were observed across all groups, suggesting that the memory impairments were not due to changes in general activity or anxiety levels. CO-exposed rats failed to habituate and could not discriminate between new and old objects, demonstrating significant memory deficits.

De Salvia M.A. Cagiano R. Carratu M.R. Di Giovanni V. Trabace L. Cuomo V.	Irreversible impairment of active avoidance behavior in rats prenatally exposed to mild concentrations of carbon monoxide.		Active avoidance conditioning	%CAR at 3 months for 75ppm = 50%; at 18 months = 80%; at 18 months reacquisition = 90%; Both age groups (3 and 18 months old) from the 150 ppm CO exposure showed significant impairments in learning the active avoidance task compared to controls. No significant effects were found on intertrial activity or escape latencies, suggesting that the learning impairments were not due to general activity or motivational issues. Demonstrated significant deficits in the reacquisition phase, indicating long-term cognitive impacts of prenatal CO exposure; CAR at 3 months = 10%; CAR at 18 months = 25%; reacquisition at 18 months = 75%
Packianathan S. Cain C.D. Longo L.D.	Ornithine decarboxylase activity and polyamine concentrations in fetal rat brain: Response to chronic hypoxic-hypoxia and/or carbon monoxide-hypoxia.	Fetal weight, brain ornithine decarboxylase activity and polyamine levels		Ornithine Decarboxylase (ODC) Activity: No consistent pattern of ODC activity changes was observed, suggesting that ODC might not be a reliable marker for chronic hypoxia. Polyamine Concentrations: Polyamine levels varied, with significant changes occurring under certain conditions of hypoxia, especially when combined with CO exposure. There was a significant decrease in fetal weight in the groups exposed to severe hypoxia (10.5% O2) and in groups where hypoxia was combined with CO (notably at 200 ppm).

B

Storm J.E. s J.J. L.D.	alterations in cerebellar GABA content, GABA uptake and morphology following exposure to carbon monoxide early in	GABA protein levels; cerebellar morphology; high affinity GABAergic and glutaminergic uptake		Exposure to 300 ppm carbon monoxide (CO) from conception to 10 days post-birth in rats led to significant reductions in cerebellar weight, impaired GABAergic neuronal maturation, and decreased 3H-GABA uptake, with fewer cerebellar fissures observed at 21 days
Mactutus C.F. Fechter L.D.	Moderate prenatal carbon monoxide exposure produces persistent, and apparently permanent, memory deficits in rats.		2 way active avoidance paradigm	At 120 days old: •Acquisition: Both the CO-exposed and control rats learned the task equally well during the initial training session, indicating that prenatal CO exposure did not impair the initial learning ability. •Retention: CO-exposed rats demonstrated significantly poorer memory retention compared to controls. They showed no significant improvement in performance over the previous day, whereas control subjects did show significant retention. At 300 days old: •Acquisition: CO-exposed aged rats showed a slower rate of learning the avoidance response compared to air-exposed controls, indicating that the memory impairment became more pronounced with age. •Retention: Like their younger counterparts, aged CO-exposed rats displayed significant deficits in memory retention. They began the reacquisition phase with performance levels similar to their initial learning, unlike controls who showed improved performance, suggesting a lack of memory “savings."

Daughtrey W.C. Norton S.	Morphological damage to the premature fetal rat brain after acute carbon monoxide exposure.	Central nervous system tissue histology; capillary stain		Extensive morphological damages were observed particularly in the ventral germinal matrix overlying the developing caudate nucleus of the CO-exposed fetuses. This region showed significant hemorrhagic infarcts and necrosis. The dorsal germinal matrix of the neopallium did not show damages from CO exposure.
Garvey D.J. Longo L.D.	Chronic low level maternal carbon monoxide exposure and fetal growth and development.	brain protein and DNA concentrations and concentrations of serotonin and norepinephrine		Hematology and blood gases were analyzed. The study noted no significant hematologic response to CO exposure in low and moderate groups but did show altered hemoglobin and hematocrit levels in the low oxygen group.
Venditti C.C. Casselman R. Smith G.N.	Effects of chronic carbon monoxide exposure on fetal growth and development in mice.	brain CO levels and weight		No significant effects on fetal or placental weights until 400ppm. At 400pm, decreased fetal mass and reduced litter size, indicating teratogenic or toxic effects. Significant increases in early and late gestational deaths (EGD and LGD). High levels of carboxyhemoglobin suggesting potential hypoxic conditions. Increased CO accumulation in maternal organs like the heart, liver, and kidneys as well as in the placenta, indicating systemic effects of high-level CO exposure.

B

Trentini J.F. O'Neill J.T. Poluch S. Juliano S.L.	Prenatal carbon monoxide impairs migration of interneurons into the cerebral cortex.	somatosensory cortex histology; BrdU immunohistochemistry and birth dating; cGMP extracted via ELISA and VASP via western blot; neuronal migration assays		Maintained normal fetal and placental weights. Elevated carboxyhemoglobin levels without clinical toxicity. Prenatal CO resulted in a thinner cerebral wall at P4 and altered the distribution of cells born on E12 and E16; Prenatal CO exposure did not alter the laminar organization of MAP2- or CAMKIIa-positive neurons; Mice were tested for sensorimotor gating using the prepulse inhibition test. Control mice reliably inhibited their response by approximately 20% however treated mice did not possess significant pre-pulse inhibition and consistently reacted strongly to the startle stimulus; CO treated pups (P0) had significantly reduced cGMP levels compared to controls (0.4 0.1 vs. 0.85 0.153 pg/mL); CO treated pups had significantly reduced phospho-VASP/VASP ratios compared to controls (1.1 0.3 vs. 1.8 0.1); Control pups had a paw withdrawal response latency of 32.7 +_ 1.8 s. Treated mice had significantly lower nociceptive thresholds, with a mean response of 22.6 +- 1.5 s

Lopez I.A. Acuna D. Beltran-Parrazal L. Lopez I.E. Amarnani A. Cortes M. Edmond J.	Evidence for oxidative stress in the developing cerebellum of the rat after chronic mild carbon monoxide exposure (0.0025% in air).	immunocytochemical analyses of oxidative stress markers, and structural and functional proteins in the cerebellar cortex; qPCR for inducible, neuronal, and endothelial nitric oxide synthases		Markers like superoxide dismutase (SOD1 and SOD2), heme oxygenase-1 (HO-1), and inducible nitric oxide synthase (iNOS) were elevated in CO-exposed groups, indicating increased oxidative stress. There was a reduction in neurofilaments, synapsin-1, and glutamic acid decarboxylase (GAD), suggesting impaired neuronal function and development. Prenatal exposure appeared to have more severe effects compared to postnatal exposure, indicating a critical vulnerability of the developing brain to CO exposure during gestation.
Beltran-Parrazal L. Acuna D. Ngan A.M. Kim E. Ngan A. Kawakami K. Edmond J. Lopez I.A.	Neuroglobin, cytoglobin, and transcriptional profiling of hypoxia-related genes in the rat cerebellum after prenatal chronic very mild carbon monoxide exposure (25 ppm).	immunohistochemistry and immunoreactivity for Ngb and Cygb antibodies in the cerebellum; Quantitative real-time RTâ€“PCR		Group A (Prenatal CO Exposure Only). No changes to the mRNA expression for neuroglobin (Ngb) and cytoglobin (Cygb). Significant increase in Cygb protein immunoreactivity. Group B (Prenatal and Postnatal CO Exposure). Similar to Group A in terms of Ngb and Cygb mRNA and protein changes. Additional gene expression analysis shows some genes related to oxidative stress were upregulated. Group C (Postnatal CO Exposure Only): Similar findings for Ngb and Cygb as in Groups A and B. Certain genes related to growth and oxidative stress were distinctly up- or downregulated compared to controls. Control Group (Group D): Used for baseline comparisons; no CO exposure.

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V. Lorusso L. Flace P. …mo F. …ni R. Auteri P. Bosco L. Cagiano R.	Effects of prenatal exposure to the CB-1 receptor agonist WIN 55212-2 or CO on the GABAergic neuronal systems of rat cerebellar cortex.	immunostaining of the cerebellar cortex for GAD-65/67 and GABA; morphometry of the cerebellar cortex		Neuroglobin (Ngb) was found to be expressed in the spiral ganglia neurons (SGNs), spiral ligament fibrocytes, and supporting cells of the organ of Corti in normal adult rat cochlea. Chronic exposure to very mild carbon monoxide (CO) (25 ppm) during prenatal and postnatal periods led to a significant decrease in Ngb immunoreactivity and mRNA expression in SGNs and spiral ligament fibrocytes, but not in supporting cells. Cytochrome-C (Cyt-C) immunoreactivity also decreased in the same regions following CO exposure. Postnatal CO exposure alone did not result in significant changes in Ngb or Cyt-C expression.
V. Lorusso L. Coluccia A. Tarullo A.	decarboxylase and GABA immunoreactivities in the cerebellar cortex of adult rat after prenatal exposure to a low concentration of	immunostaining of the cerebellar cortex for GAD-65/67 and GABA; morphometry of the cerebellar cortex		Significant reduction in GAD 65/67 and GABA immunoreactivity in the molecular and Purkinje neuron layers, suggesting a potential impact on GABAergic neurons despite no visible changes in the general architecture of the cerebellar cortex

Vaccari A. Ruiu S. Saba P. Fa M. Cagiano R. Coluccia A. Mereu G. Steardo L. Tattoli M. Trabace L.	Prenatal low-level exposure to CO alters postnatal development of hippocampal nitric oxide synthase and haem-oxygenase activities in rats.	hippocampus incubated in the presence of [14C]arginine (for nNOS) and [14C]haem (for HO-2) and metabolites measured		Control: Hippocampal Nitric Oxide Synthase activity: 15d old: 52 pmol/mg protein per 15 min; 30d: 55 pmol/mg protein per 15 min; 90 days old: 69 pmol/mg protein per 15 min; Hippocampal Haem-Oxygenase (HO2) activity: 15d: 1054 pmol/mg protein per hour; 30d 1173 pmol/mg protein per hour; 90d: 1082 pmol/mg protein per hour; CO exposure: Hippocampal Nitric Oxide Synthase activity: 15d old: 31 pmol/mg protein per 15 min; 30d: 33 pmol/mg protein per 15 min; 90 days old: 35 pmol/mg protein per 15 min; Hippocampal Haem-Oxygenase (HO2) activity: 15d: 945 pmol/mg protein per hour; 30d 797 pmol/mg protein per hour; 90d: 805 pmol/mg protein per hour
Morris GL Curtis SE Simon J	Perinatal piglets under sublethal concentrations of atmospheric carbon monoxide.	histopathological examination of representative tissues from the brain	negative-geotaxis tests; open field test; time to first nursing	Perinatal exposure to atmospheric carbon monoxide at sublethal concentrations led to delayed nursing behavior in piglets. Piglets exposed to carbon monoxide at 250 ppm took longer to nurse for the first time. Exposure to carbon monoxide also resulted in compromised performance in behavior tests evaluating the piglets' ability to orient, maneuver, and investigate their environment.
Storm JE Fechter LD	Prenatal carbon monoxide exposure differentially affects postnatal weight and monoamine concentration of	Monoamine neurotransmitters (NE, 5HT, and DA) assayed via concentrations of metabolites; regional brain weights of the medulla, neocortex, hippocampus and		Monoamine neurotransmitters (NE, 5HT, and DA) assayed via concentrations of metabolites; regional brain weights of the medulla, neocortex, hippocampus and cerebellum

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Storm JE Fechter LD	postnatal ontogeny of cerebellar norepinephrine content following chronic prenatal	cerebellar and cortical weight; NE concentration in the cerebellum and the cortex using HPLC with electrochemical detection		cerebellar and cortical weight; NE concentration in the cerebellum and the cortex using HPLC with electrochemical detection
Mactutus CF Fechter LD	Prenatal exposure to carbon monoxide: learning and memory deficits.		two-way conditioned avoidance response	Exposure to CO in utero exhibited significant impairments in learning and memory in a two-way avoidance task. The observed deficits were found to be associative, with no evidence of alterations in nonassociative or motivational processes. Reacquisition of the avoidance task was also adversely affected, showing impaired retention of learned behavior in CO-exposed offspring compared to controls.
Daughtrey WC Norton S	Caudate morphology and behavior of rats exposed to carbon monoxide in utero.	Golgi stain and toluidine blue staining of the caudate nucleus	surface righting; auditory startle; eye opening; reflex suspension; open field; residential maze; swimming ability; exploratory hole board; residential maze with drug challenge	Rats exposed to CO in utero showed significant malformations in the caudate nucleus, including ectopic caudate tissue projecting into the lateral ventricles. The incidence of ectopias increased with longer exposure to CO. CO exposure led to a reduction in dendritic branching of Golgi type II neurons in the caudate nucleus, observed at 1 day of age, but this effect was not permanent and normalized by 7 days of age. No significant behavioral deficits were observed in tests assessing motor function and locomotor activity in CO-exposed rats. No reduction in litter size or growth rate of offspring during the preweaning period. No consistent long-term behavioral changes in CO-exposed rats

Ginsberg MD Myers RE	Fetal brain injury after maternal carbon monoxide intoxication. Clinical and neuropathologic aspects.	brain histology	reflex grasping; suck response	Control: Gestational day 19: Putrescine: 3.7 + 0.4^ (24) Spermidine: 11 +_ 1^ (23) Spermine: 7.3 + 0.7^(24); Gestational day 21: Putrescine: 3.7 ± 0.3^ (26); Spermidine: 13 _+ 1 ^(23); spermine 8.9 +_ 0.8^ (25); 100ppm: G19: p: 4.3 -+ 0.5 ^(15) ; s: 12 ± 1 ^(14); s: 7.8 +- 0.8 ^(15) ; g21: p: 3.3 _+ 0.4 (15) s: 11 +_ 1^(14); 200ppm: 4.2_+0.5 ^ (11) ; 12 +1^11; 9 _+1^11; 3.5+0.5 (11) ; 11 +1^11

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Protocol written for systematic review

Study protocol for a preclinical review of the neurotoxic effects of sub-acute and chronic carbon monoxide exposure

Alessa Weiler * Robert Dickinson†

June 18, 2024

1. Rationale

Carbon monoxide (CO) is produced from incomplete combustion of fuels, as well as being an endogenous gasotransmitter. While the neurological effects of acute high-dose CO poisoning is well documented, the neurotoxic impact of chronic low-level exposure are less understood and potentially significant. Chronic exposure to low levels of CO, such as from malfunctioning heating devices, is posited to cause subtle yet potentially significant effects on the brain (Harper and Croft-Baker, 2004). Symptoms reported from anecdotal evidence and case studies include headaches, malaise and cognitive impairments, which may improve after ceasing the exposure (Myers, Snyder and Ermhoff, 1965; Thom, Thaber et al., 1995; Ryan, 1990; Sherral, Devine et al., 2002; Dolan et al., 1987; Bendell et al., 2019). The world health organisation (WHO) has set guidelines to prevent COHb levels from rising above 2.5%, but there is still a lack of data on the long-term neurological effects of low-level exposure within these limits (WHO Environmental Health Criteria 213).

Despite the known dangers of high-level CO exposure, very low-level CO has been investigated for its potential therapeutic benefits in diseases such as Parkinsons (Gompertz et al., 2023; Rose et al., preprint) or lung injury (SOURCE), such as its anti-inflammatory and cytoprotective effects. However this focus on the therapeutic aspects contrasts with the potential nefarious effects of low-level CO exposure on the brain. There is a critical need to assess these risks, especially given that low-level CO treatment is already clinical trials (Rosas et al., 2018). Our systematic review aims to understand whether there is evidence of brain injury due to exposure to sub-acute (sub-lethal) or repeated CO exposure .

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2. Methods

2.1. PICO framework

Population Preclinical studies in adult or juvenile animals (mammals)

Intervention Exposure to low or sub-lethal concentrations of carbon monoxide with either single or repeated exposure. Our aim is to determine the neurotoxic effects of low level carbon monoxide poisoning.

Outcomes Our primary outcomes are evidence of neurological injury, including histological evidence of neuronal injury or functional neurological impairment as measured by brain changes, locomotor outcome, behavioural changes, electrophysiological alterations, neuropsychological performance and neurochemical changes. Included secondary outcomes were physiological outcomes including weight change or levels of carboxyhaemoglobin.

Literature search and study selection We will consider literature published in English. Searches will be carried out on the Ovid Medline (PubMed) and on EMBASE databases. In addition, the reference lists of potential eligible literature will be reviewed to identify any relevant papers, and a backwards and forwards citation analysis will be performed. Titles, abstracts and full text of articles will be independently screened by two reviewers. Where discrepancies are found a third reviewer will adjudicate.

Inclusion Criteria Eligible studies are peer-reviewed studies on the neurotoxic effects of low level carbon monoxide exposure in adult or juvenile animals (mammals), published in English. There are no restrictions on the year of publication, initiation and duration of treatment, or severity of injury.

Exclusion Criteria Studies will be excluded if they (1) are human studies, (2) did not investigate carbon monoxide exposure (3) lack required outcome measures, (4) are review articles, (5) are conference abstracts.

Data Extraction Data will extracted independently by two reviewers. Where discrepancies are found a third reviewer will adjudicate. If data is in numerical or text format this will be extracted manually. For studies where data is in graphical format, this will be extracted from calibrated digitised plots using a digitiser tool (eg https://automeris.io/WebPlotDigitizer/). If data is missing or incomplete the corresponding author will be contacted.

Risk of Bias Study quality and risk of bias will be assessed independently by two reviewers. Where discrepancies are found a third reviewer will adjudicate. The following criterial will be used (based on the CAMARADES checklist) (1) physiological measurements (blood pressure, heart rate, or blood gases), (2) randomisation, (3) blinding of protocol, (4) blinding

of outcome assessment, (5) sample size calculation, (6) compliance with animal welfare regulations, (7) peer-reviewed publications, (8) potential conflicts of interest and (9) controlling temperature. Each item will be scored '1', otherwise '0', giving a score of 0 (high risk of bias) to 9 (low risk of bias).

Outcomes included Physiological brain changes, functional neurologic or locomotor outcome, neuropsychological behaviour performance, levels of carboxyhaemoglobin, weight change

2.2. Analysis Plan

If sufficient studies with the appropriate quantitative data are identified we plan to carry out a meta analysis as described below. If there is insufficient data for a meta analysis then a narrative review will be provided.

2.2.1. Assessing heterogeneity and publication bias overall

Heterogeneity Potential sources of heterogeneity will be explored using the I2 statistic. The predefined potential sources of heterogeneity are animal species, length of time of exposure, study quality, sample size calculation, randomisation, blinding to assessment of outcome, temperature control and inclusion of sham group.

Publication bias Potential publication bias (missing studies) will be examined by constructing funnel plots and using Egger's regression. The effect of funnel plot asymmetry on summary effects will be quantified using the trim-and-fill method.

2.2.2. Analysis plan for a meta analysis

Individual study analysis First, for each study included, data will be extracted as described in the Data Extraction section. We will focus on all relevant outcomes that measure neurological injury. Where data on the number of individuals with primary and secondary outcomes of interest was not available, we extracted the information as it was presented (eg., we noted effect sizes, percentage changes or mean values along with their standard deviations). A pairwise random effects model meta-analysis of normalized mean differences (NMDs) between the control and treatment groups will be conducted for normalisation. The inverse variance method will be employed for weighting, the restricted maximum likelihood (REML) estimator for variance (tau^2), and the Q-profile method for the confidence interval of tau^2 and the overall effect size (tau). This approach will provide a single overall effect size and standard error (SE) for each study.

Aggregate analysis across studies These individual study effect sizes and SEs will then be aggregated in the overall random effects meta-analysis for each type of exposure assessed (low or sub-lethal concentrations of carbon monoxide), using the inverse variance method for weight, the REML estimator for tau^2, and the Q-profile method for the confidence interval

of tau^2 and tau. The heterogeneity of the neurotoxic effects across all included studies will be quantified using the heterogeneity index, I^2, and tested using the Q statistic with a significance value of $p < 0.05$. The rationale for employing this two-step procedure looking both at individual studies and aggregating studies is to provide a single estimate of efficacy for each study, thus preventing the final aggregated estimate from being biased towards studies that reported many different outcomes and/or repeated measurements.

Heterogeneity Potential sources of heterogeneity will be explored using stratified meta-analyses and meta-regression. The predefined potential sources of heterogeneity are animal species, length of exposure, study quality, sample size calculation, randomisation, blinding to the assessment of outcomes, temperature control, and inclusion of a sham group. Differences in subgroups in the stratified meta-analyses will be tested with a chi-square test.

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